

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Open Data Analysis

Course Code: DSA0502

CO Assessed: CO4

Weightage:

Course Title: Query Processing for Data Science

Assessment: Tool 2– Open Data Analysis

Total Marks:

CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Student Name: Sandra Sudevan

Section / Batch: 2024

Faculty In-charge: Dr. B.Shyamala Gowri

Reg. No.: 192424422

Date of Submission: 24/08/26

Project Mode: Individual Submission

Code of Conduct

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandra Sudevan

8. Select any **open tourism dataset** containing tourist arrivals, destinations, hotel occupancy, travel expenditure, or visitor information. Explore and analyze the data using grouping, filtering, correlation, outlier detection, and time-series techniques. Present the findings through charts, maps, and interactive visualizations.

Aim

To select an open tourism dataset and analyze tourist arrivals, destinations, hotel occupancy, and travel expenditure using different data-analysis techniques.

Dataset Used

For this program, we use a **Tourism Dataset** containing:

- **Year** – Year of tourism activity
- **Destination** – Tourist destination/country
- **Tourist Arrivals** – Number of tourists visiting
- **Hotel Occupancy** – Percentage of hotel rooms occupied
- **Expenditure** – Tourist spending

Techniques Used

1. Grouping

The data is grouped according to **destination** to calculate the total number of tourist arrivals for each destination.

Purpose:

To identify which destination receives the most tourists.

2. Filtering

Records having tourist arrivals **above the average** are selected.

Purpose:

To identify destinations and years with high tourist activity.

3. Correlation Analysis

Correlation is calculated between:

- Tourist arrivals
- Hotel occupancy
- Expenditure

Purpose:

To determine whether these tourism factors are related to each other.

For example, a positive correlation between tourist arrivals and hotel occupancy indicates that hotel occupancy tends to increase when tourist arrivals increase.

4. Outlier Detection

The **IQR (Interquartile Range)** method is used to detect unusual tourist-arrival values.

Purpose:

To identify years or destinations with exceptionally high or low tourist arrivals.

5. Time-Series Analysis

Tourist arrivals are grouped year-wise and displayed using a **line graph**.

Purpose:

To understand how tourism changes over time.

6. Visualization

Different charts are created:

- **Bar chart** – compares tourist arrivals by destination.
- **Line chart** – shows tourism trends over the years.
- **Heatmap** – displays correlations between variables.
- **Box plot** – identifies outliers.
- **Scatter plot** – shows the relationship between tourist arrivals and hotel occupancy.
- **Interactive Plotly chart** – allows users to explore the tourism data interactively.

Expected Findings

From the analysis, we can determine:

- The destination with the highest tourist arrivals.
- The year with the highest tourism activity.
- Whether hotel occupancy increases with tourist arrivals.
- The relationship between tourism expenditure and arrivals.
- Whether any unusual/outlier observations exist.
- The overall tourism trend over the years.

Conclusion

The tourism dataset can be effectively explored using Python and Pandas. Grouping and filtering help identify important tourism patterns, correlation reveals relationships between variables, outlier detection identifies unusual observations, and time-series analysis shows changes in tourism over the years. Charts and interactive visualizations make the results easier to understand.

Figure 1

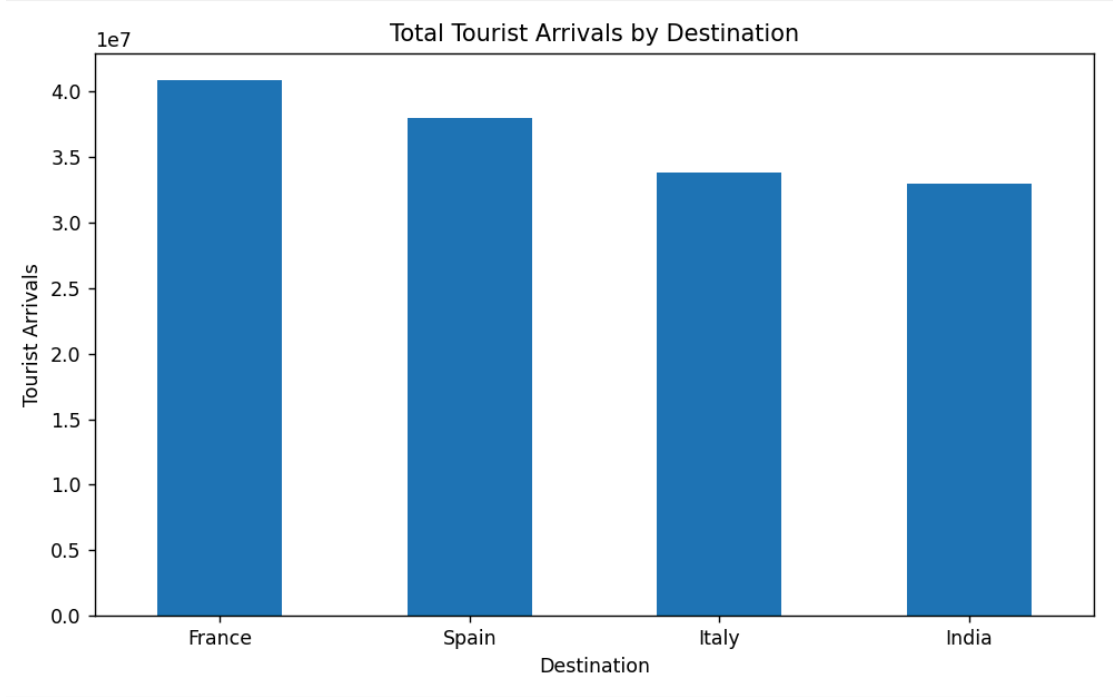


Figure 2

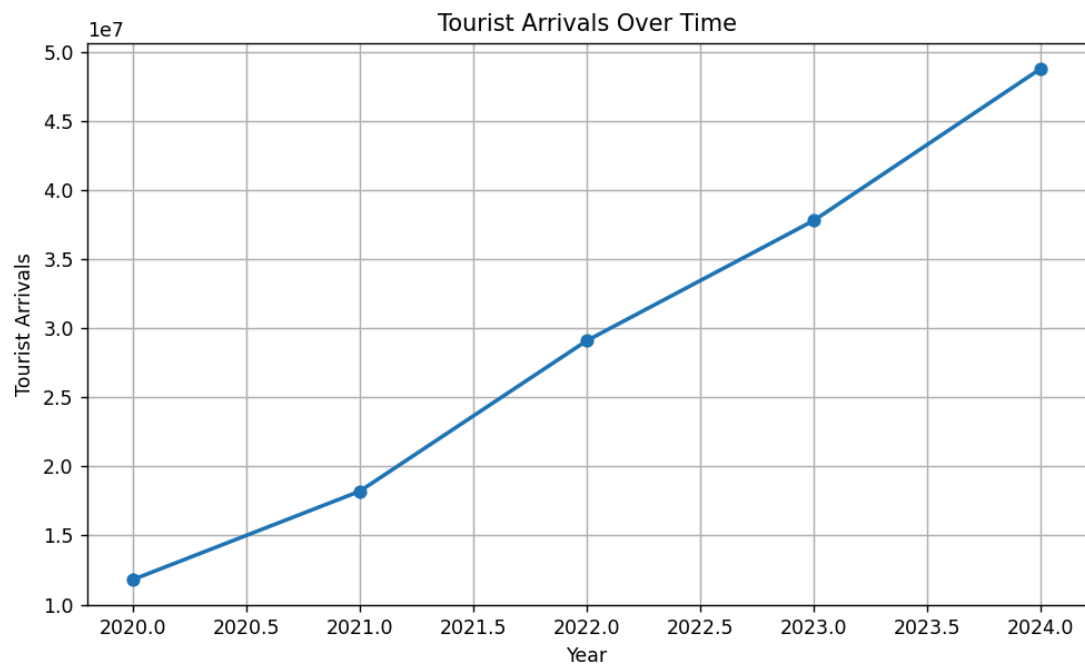


Figure 3

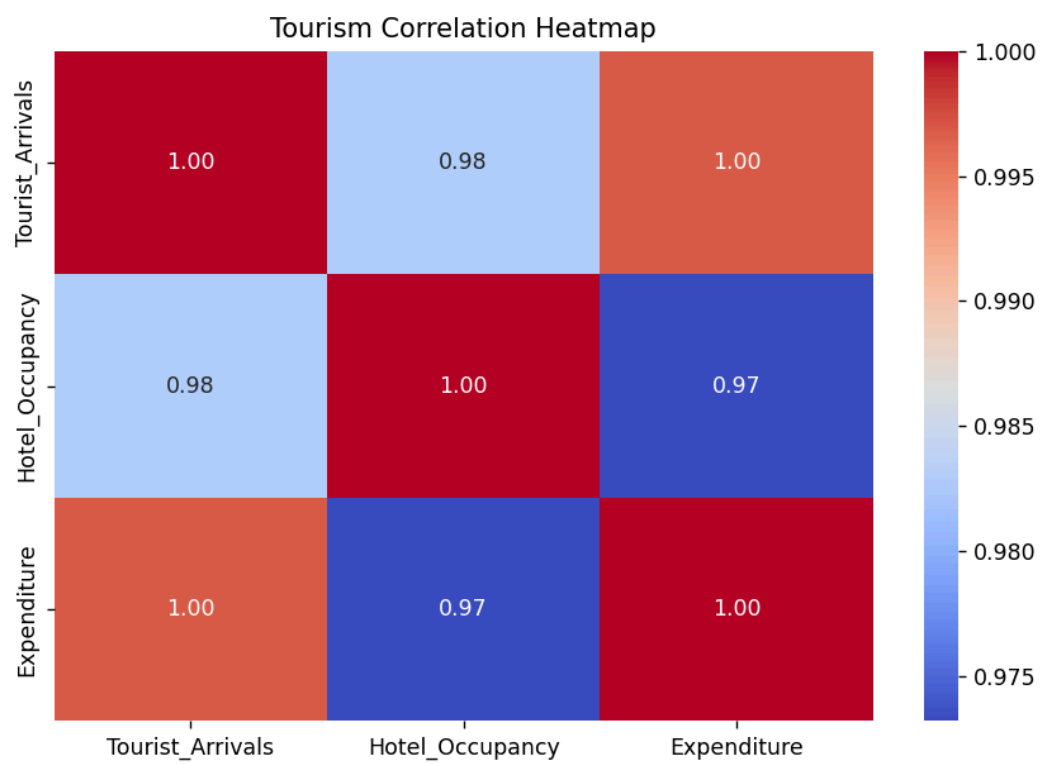


Figure 4

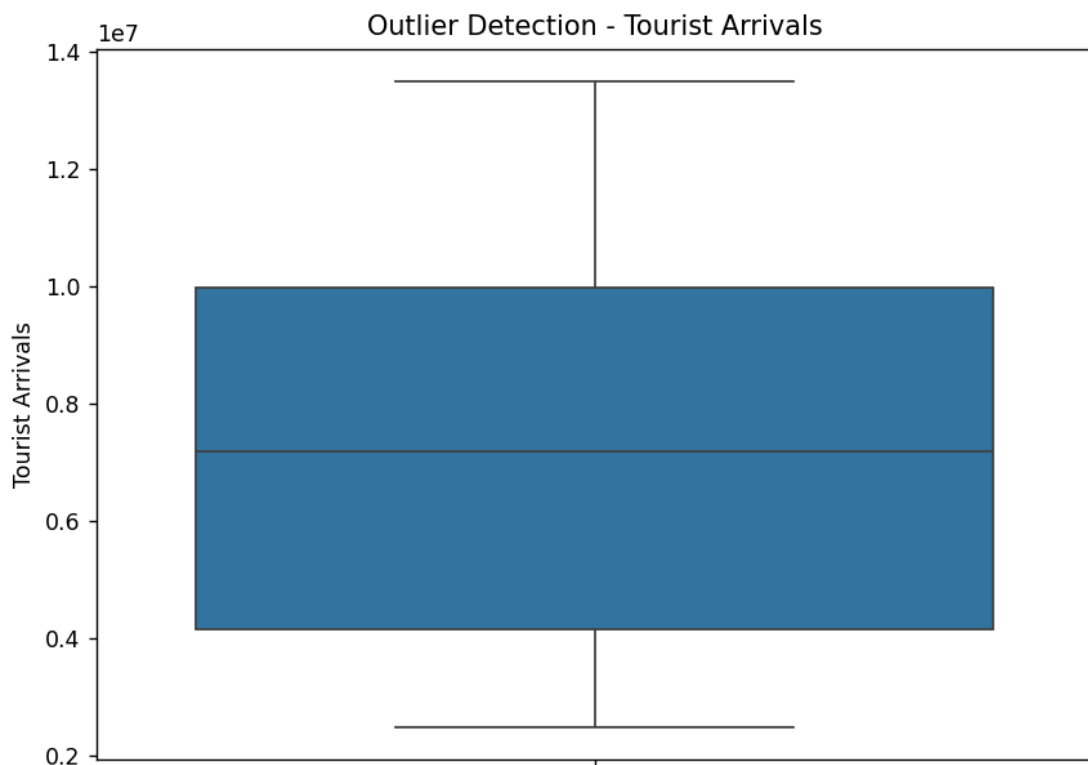
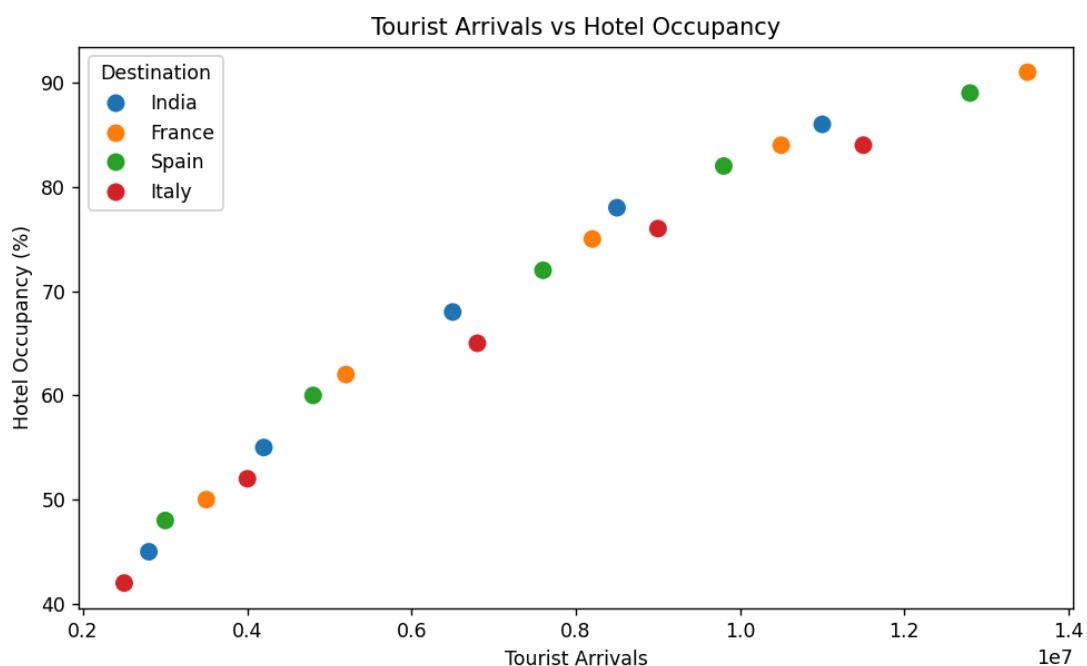


Figure 5



10. Select any **open social media or web dataset** containing posts, comments, likes, shares, hashtags, keywords, or user activity. Analyze patterns, create groups, identify relationships and unusual observations, perform time-based analysis, and present the results using charts, word-based visualizations, interactive dashboards, and other suitable presentation tools.

Aim

To select an open social media dataset and analyze posts, likes, comments, shares, hashtags, keywords, and user activity using different data-analysis techniques.

Dataset Used

The dataset contains social media information such as:

- **Date** – Date of the post
- **Platform** – Instagram, YouTube, Facebook
- **Category** – Travel, Technology, Food
- **Likes** – Number of likes
- **Comments** – Number of comments
- **Shares** – Number of shares
- **Hashtag** – Hashtag used in the post
- **Keyword** – Main keyword of the post

Techniques Used

1. Grouping

Posts are grouped based on **platform and category**.

For example, total likes are calculated for Instagram, YouTube, and Facebook.

Purpose:

To identify which platform receives the highest engagement.

2. Filtering

Posts having **likes greater than the average number of likes** are selected.

Purpose:

To identify highly engaging or popular posts.

3. Correlation Analysis

Correlation is calculated between:

- Likes
- Comments
- Shares

Purpose:

To identify relationships between different engagement metrics.

For example, a positive correlation between likes and shares means posts with more likes generally tend to receive more shares.

4. Outlier Detection

The **IQR method** is used to detect posts with unusually high or low numbers of likes.

Purpose:

To identify unusual or exceptionally popular posts.

5. Time-Based Analysis

The posts are grouped by **month** and total likes are calculated.

Purpose:

To identify which month has the highest social-media activity.

6. Hashtag Analysis

The frequency of each hashtag is calculated.

Purpose:

To identify the most frequently used or popular hashtags.

7. Keyword / Word Visualization

The keywords from the posts are analyzed and displayed as a **word-based visualization**.

Purpose:

To identify the most common topics discussed in the dataset.

Visualizations

The program creates:

1. **Bar chart** – Total likes by platform
2. **Line chart** – Monthly social-media activity
3. **Correlation heatmap** – Relationship between likes, comments, and shares
4. **Box plot** – Outlier detection
5. **Scatter plot** – Likes vs. shares
6. **Keyword visualization** – Frequently used words
7. **Interactive Plotly chart** – Interactive engagement analysis

Expected Findings

The analysis can identify:

- Which platform has the highest engagement.
- Which category receives the most attention.
- Which posts have unusually high engagement.
- Whether likes, comments, and shares are related.
- Which month has the highest activity.
- Which hashtags and keywords are most popular.

Conclusion

The social media dataset can be effectively analyzed using Python. Grouping and filtering help identify popular platforms and posts, correlation shows relationships between engagement measures, outlier detection identifies unusual posts, and time-based analysis reveals activity trends. Charts and interactive visualizations make the results easier to understand.

